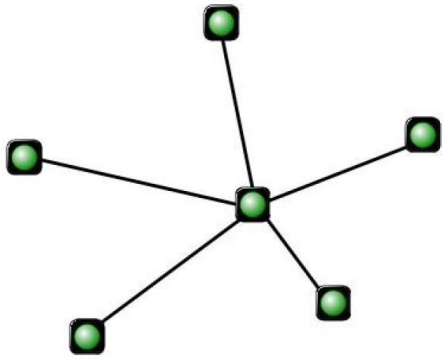
Roboteam

DTC Operator Interface Guide

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Training & Field Reference for the NETNode Mesh IP Radio Web Interface



Item	Details
Document	DTC Interface User Guide (English edition)
Intended audience	Operators, field technicians and trainers
Companion document	Full manufacturer user manual (B.A. Microwaves / DTC)
Classification	Proprietary Information – Commercially Classified

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1. About This Guide

1.1. Purpose

The "Terminator" project combines several advanced communication modules. Its central and most critical operational backbone is the RF communication infrastructure supplied by DTC. To succeed in every phase of an operation, from mission planning, through real-time management, to post-mission analysis and debriefing, operators must have full command and a solid understanding of the DTC radio web interface.

The interface gives operators a broad technical and operational picture. It allows them to:

- View live data from the **Mesh network** in real time.
- Run a **spectral analysis** of the operating frequencies.
- Monitor the **effective communication range** of each link.
- Inspect the **relay matrix**, showing how well every node relays to every other node.

With this information, commanders and operators in the field can make informed decisions, optimize network performance and maintain operational continuity. Skilled use of the interface directly improves command and control and gives a clear advantage when managing the systems in each operating configuration.

> NOTE
> This guide covers only the most important functions and settings. For complete details, refer to the full manufacturer user manual (B.A. Microwaves), which is supplied alongside this document.

1.2. Conventions Used in This Guide

Convention	Meaning
Apps > Tactical Display	A menu path in the interface. Click each item in order.
IMPORTANT	Information you must follow for the system to work correctly.
CAUTION	An action that can cause loss of communication if done incorrectly.
TIP	Practical advice that makes the task easier.

2. System Overview

2.1. Network Components

Every radio in the system is a node in a single Mesh network. The table below lists the components you will see in the interface.

Component	Role in the network
Ground Control Station (GCS) – Phase-5	Central ground station. In Mesh configuration it provides a view of every node in the network.
Operator Box (BLU-SDR)	The operator's radio box. There is one per platform (Platform I and Platform II).
ROCU-10	The operator control unit. It connects to the Operator Box by cable and its built-in web browser is used to open the interface.
Vehicles (Bison 1 and Bison 2)	The two unmanned ground vehicles, each carrying a DTC radio.
Main relay antenna	Ground relay antenna mounted on a trailer or tripod mast.
Relay drone	Airborne relay. Raising or repositioning it extends and improves coverage to the vehicles.

2.2. Operating Configurations and IP Addresses

The interface can be opened in two configurations. The IP address you enter determines which configuration you are working in.

Configuration	IP address	What you see
Tactical (BLU-SDR)	Platform I: 192.168.120.131 Platform II: 192.168.121.131	The Operator Box of the platform your ROCU-10 is connected to. Use the address that matches that platform.
Mesh (Phase-5)	192.168.168.133 (same for both platforms)	Data from every node in the network: GCS, both vehicles, main relay antenna and relay drone. This allows analysis of the complete system.

> TIP
>
> Use the **Mesh (Phase-5)** configuration when you need the full picture of the network, for example during setup, frequency changes and troubleshooting.

3. Connecting and Logging In

3.1. Before You Start

- The ROCU-10 is connected by cable to the Operator Box (the normal wired connection).
- The Operator Box is powered on.
- You know which configuration you need and its IP address (see Section 2.2).

3.2. Login Procedure

1. Connect the ROCU-10 to the Operator Box with the wired cable.
2. Open the built-in web browser on the ROCU-10.
3. Type the IP address for your configuration into the address bar and press Enter.
4. When the correct address is entered, the login screen opens.
5. Leave the **Username** field empty. In the **Password** field only, enter **Hatal**.
6. If that password is not accepted, try **Eastwood**.
7. Click the login button. The Home screen (Dashboard) opens.



Figure 1. Interface login screen

> TIP
>
> If the login screen does not appear, check the cable between the ROCU-10 and the Operator Box and confirm that the IP address matches your configuration.

4. Home Screen (Dashboard)

After you log in, the interface opens on the Home screen. It has three main areas: the top status bar, the unit and mesh information panel, and the navigation menu on the left.

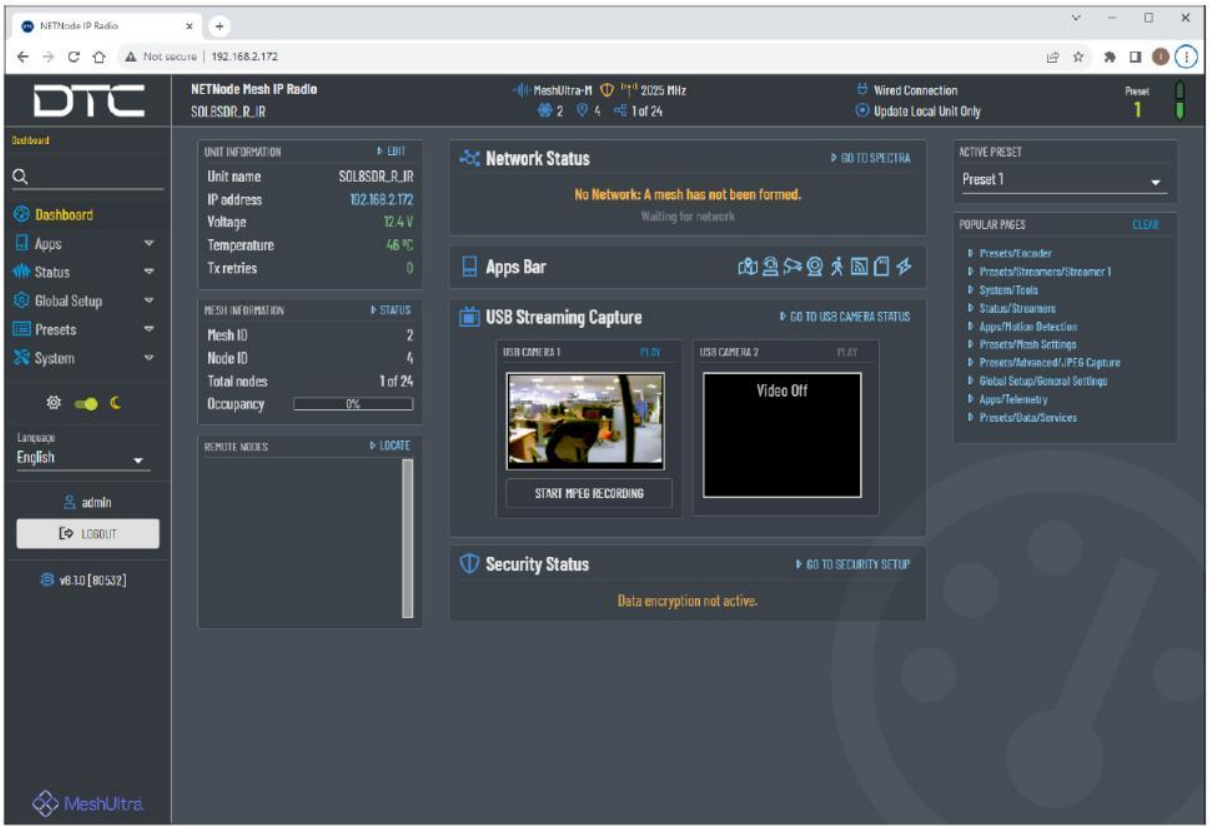


Figure 2. Home screen overview

4.1. Top Status Bar

A bar of key parameters is shown at the top of every page.

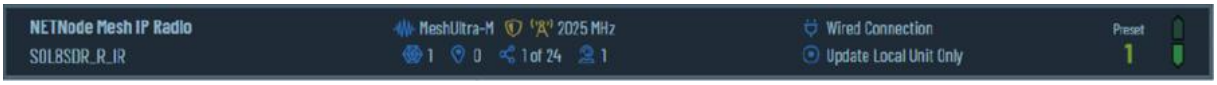


Figure 3. Top status bar

Element	Description
Connected unit name	Shown on the left, under "NETNode Mesh IP Radio". This is the radio you are logged in to.
Configured frequency	The operating frequency currently set on the radio (for example, 2025 MHz).
Mesh Zone	The Mesh network the radio belongs to.
Node ID	The radio's identifier within the Mesh network.
Wired connection indicator	Shows that you are connected to the radio over the wired connection.
Preset	The preset currently active on the radio.
Connection status bar	Far right. A red indication means a connection failure.

4.2. Unit and Mesh Information Panel

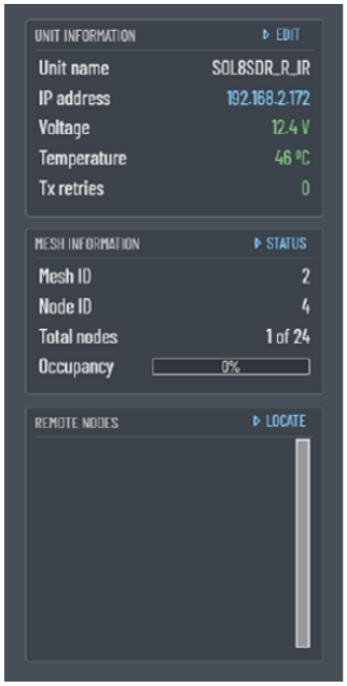


Figure 4. Unit and mesh information panel

Section	Field	Description
Unit Information	Unit name	Name of the radio.
Unit Information	IP address	IP address of the radio.
Unit Information	Voltage / Temperature	Supply voltage and unit temperature.
Unit Information	Tx retries	Number of transmission retries.
Mesh Information	Mesh ID / Node ID	Network and node identifiers.
Mesh Information	Total nodes	Number of nodes currently in the network, out of the maximum (for example, 1 of 24).
Mesh Information	Occupancy	Bandwidth load of this node, in percent.
Remote Nodes	Node list	The nodes participating in the Mesh relay network.

4.3. Navigation Menu



Figure 5. Navigation menu

Menu item	What it contains
Search bar	Find a specific setting or function quickly.
Dashboard	Returns to the Home screen.
Apps	Operational functions of the system, including Tactical Display and Spectrum Scan.
Status	Live system status: SNR values, the Mesh quality matrix, CPU usage and relay quality.
Global Setup	Settings that apply across the network, including General Settings and Update all nodes.
Presets	Radio settings saved per preset, including Mesh Settings and the operating frequency.
System	System tools and administration.

5. Apps

5.1. Tactical Display

> IMPORTANT
>
> The Tactical Display is currently the **most important indicator of communication quality**. Use it to decide when a vehicle must stop advancing.

To open: Apps > Tactical Display

The map shows every radio currently connected to the relay network, including Bison 1 and 2, the main relay antenna, the relay drone and the BLU-SDR Operator Boxes. Each radio appears as a tag labeled with its name. You can drag the tags freely around the map to spread them out for easier reading; the position of a tag on the map is not critical.

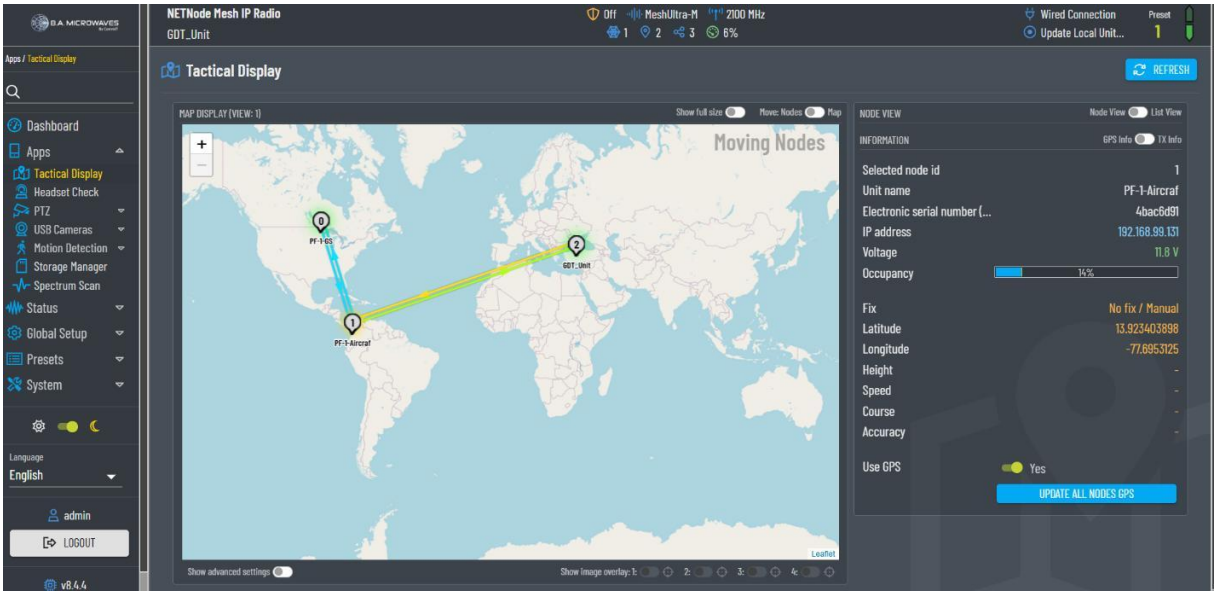


Figure 6. Tactical Display

5.1.1. Reading the link arrows

Each link between two radios is drawn as arrows, and the color of each arrow shows the signal quality:

- **Outgoing arrow:** quality of the signal being transmitted.
- **Incoming arrow:** quality of the signal being received.

When link quality changes (gets better or worse), the arrow color changes accordingly. Always aim to operate in the **blue to green** range.

5.1.2. Signal quality scale

Level	Color	Link quality	What to do
10 – 12	Blue	Excellent	Normal operation.
7 – 9	Green	Good	Normal operation.
5 – 6	Yellow-green / Yellow	Fair, link degrading	Monitor closely and prepare to act.
4	Orange	Significant attenuation	Act now: stop the advance and improve the relay.
1 – 3	Red	Poor, approaching link loss	Act immediately.
0	Dark red	No usable link	Recover the link.

> CAUTION
>
> When an arrow turns **orange**, it is already an indication of significant signal attenuation. Do not continue advancing the vehicle.

5.1.3. When link quality drops

1. Stop the vehicle's advance.
2. If needed, move the vehicle that is approaching link loss back toward better coverage.
3. Raise or reposition the relay drone.
4. On this screen, confirm that the main relay antenna (trailer/tripod) has a good link to the relay drone, and that the drone has a good link to the vehicle.

5.2. Spectrum Scan

Run a spectrum scan when you arrive in a new area of operations, to find the frequency range with the least interference.

To open: Apps > Spectrum Scan

> CAUTION
>
> Run a spectrum scan **only when the vehicles are close to the operator**, as part of pre-mission preparation. **Never run it during an active operation.**
>
> When a radio starts a scan, communication drops momentarily. If a vehicle is far away, it may fail to reconnect properly.

5.2.1. Running the scan

1. Go to **Apps > Spectrum Scan**.
2. Under **Scan Settings**, set the **Start frequency** and **Stop frequency** of the scan range.
3. Click **Start Scan** at the bottom of the page and wait for the chart to fill in.



Figure 7. Spectrum Scan. The red box marks a range with minor interference.

5.2.2. Reading the results

- **Spikes** (sudden, abnormal jumps in the chart) indicate interference. Avoid those frequencies.
- Choose a frequency range where interference is minor, such as the range marked in red in Figure 7.

5.3. Changing the Operating Frequency

> IMPORTANT
>
> If the frequency you choose is different from the one currently set for the system, you must set the **same frequency on every radio**: both vehicles, both Operator Boxes, the main relay antenna (tripod) and the relay drone.
>
> Any radio that is not set to the new frequency will lose communication with the network.

> TIP
>
> Before you change anything, **write down the current frequency**. If communication is worse after the change, you can return to it.

5.3.1. Option A (recommended): Update all nodes in one action

All radios must be **powered on and connected to the Mesh network** before you start.

1. Go to Global Setup > General Settings.
2. Under Update Settings, set Update all nodes to Yes.
3. Change only the frequency (Presets > Mesh Settings > Frequency). Do not change any other settings.
4. Save, then restart all radios and verify (see "After the change" below).

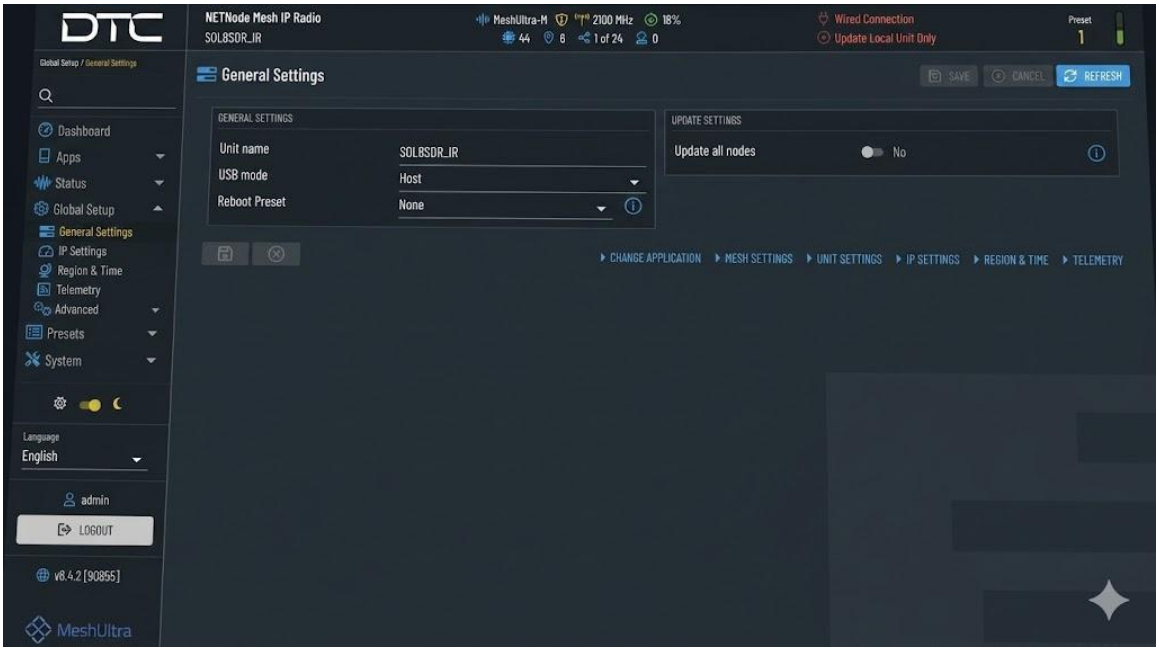


Figure 8. General Settings with the Update all nodes option

5.3.2. Option B: Configure each radio manually

1. On each radio, go to Presets > Mesh Settings.
2. In the Transmitter section, enter the new value in the Frequency field (marked in red in Figure 9).
3. Save.
4. Repeat on every radio, following the order below.

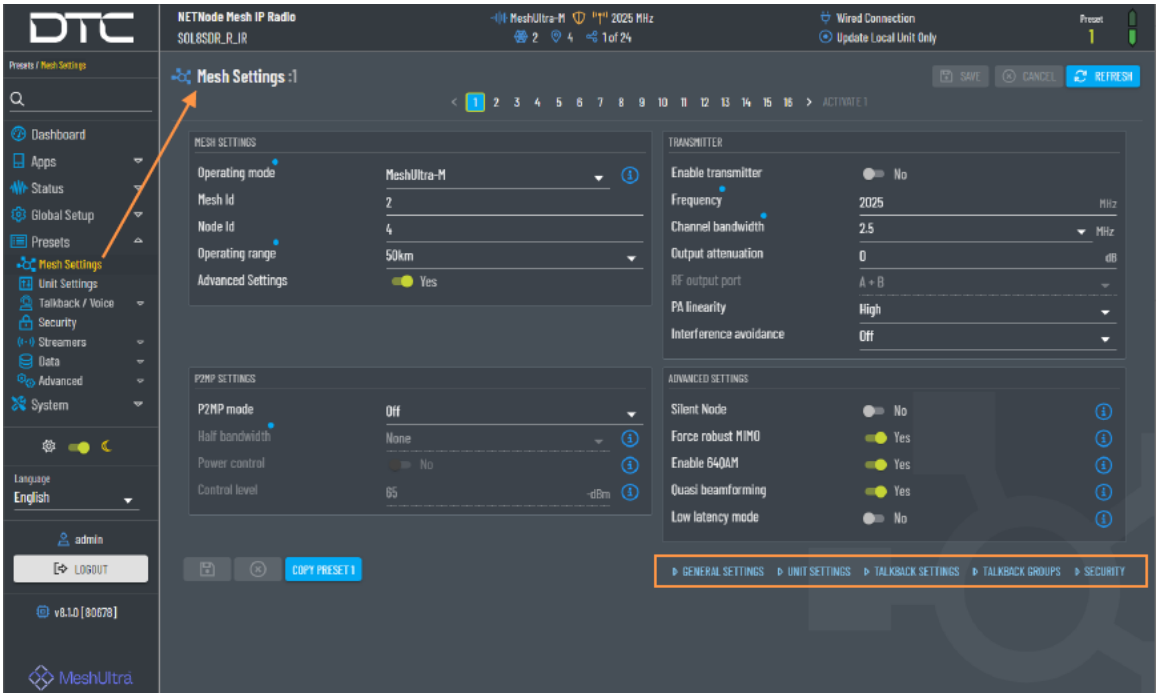


Figure 9. Mesh Settings. The Frequency field is marked in red.

Configuration order:

Order	Radio
1	Vehicles: Bison 1 and Bison 2
2	Relay drone
3	Ground Control Station (GCS)
4	Operator Boxes 1 and 2

> NOTE
> The most distant radios are configured first and the Operator Boxes last, so you do not lose your path to a radio before it has been updated.

5.3.3. After the change

1. Restart all radios.
2. Open the interface again.
3. Confirm that every radio is configured and communicating on the same network (check the Tactical Display and the Node Table).

6. Status

6.1. Node Table

To open: Status > Node Table

The Node Table lists every active station in the Mesh network and gives an essential snapshot of connection status, relay quality and the links between the stations.

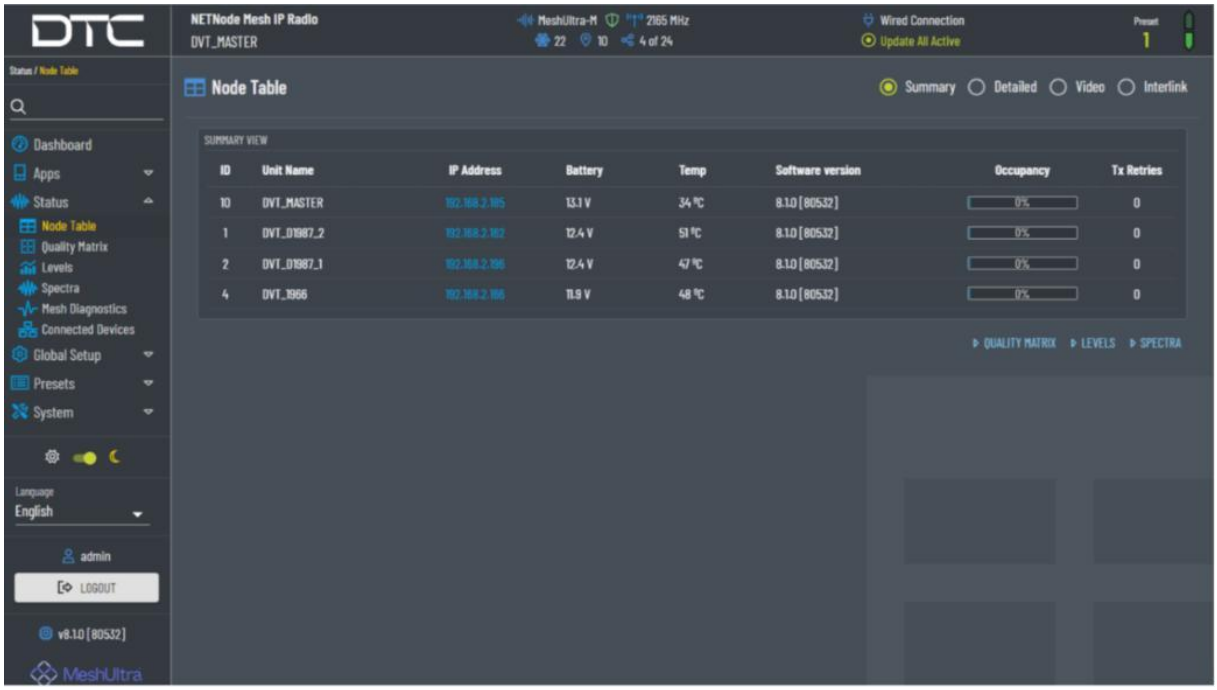


Figure 10. Node Table (Summary view)

Column	What it tells you
ID / Unit Name / IP Address	Identifies each radio in the network.
Battery	Supply voltage of the radio.
Temp	Processor temperature.
Software version	Firmware version installed on the radio.
Occupancy	Bandwidth load of the radio, in percent.
Tx Retries	Transmission retries. Use it to confirm that the radio is transmitting properly.

> CAUTION
>
> An Occupancy above **20%** is **critical**. A radio with high occupancy loads the whole network.

6.2. Quality Matrix

To open: Status > Quality Matrix

The Quality Matrix shows the transmission and relay quality between every pair of radios in the network.

Color	Meaning
Green	Good link quality.
Red	Poor link quality.
Dark red	Approaching loss of communication.

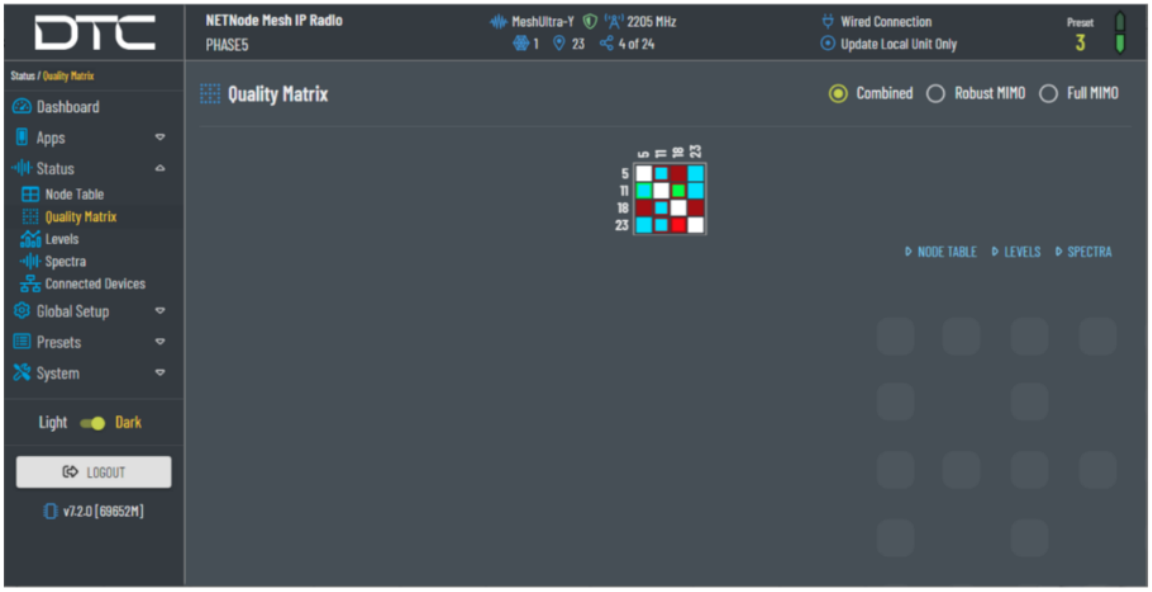


Figure 11. Quality Matrix

6.3. Levels

To open: Status > Levels. Choose the view with the selector at the top right.

View	What it shows
SNR	Signal-to-noise ratio in dB. This is a direct measure of how clean the link really is and determines whether the network can pass data without errors.
Levels	Physical received signal strength between all links in the network. It shows the level of attenuation or the raw transmit strength.

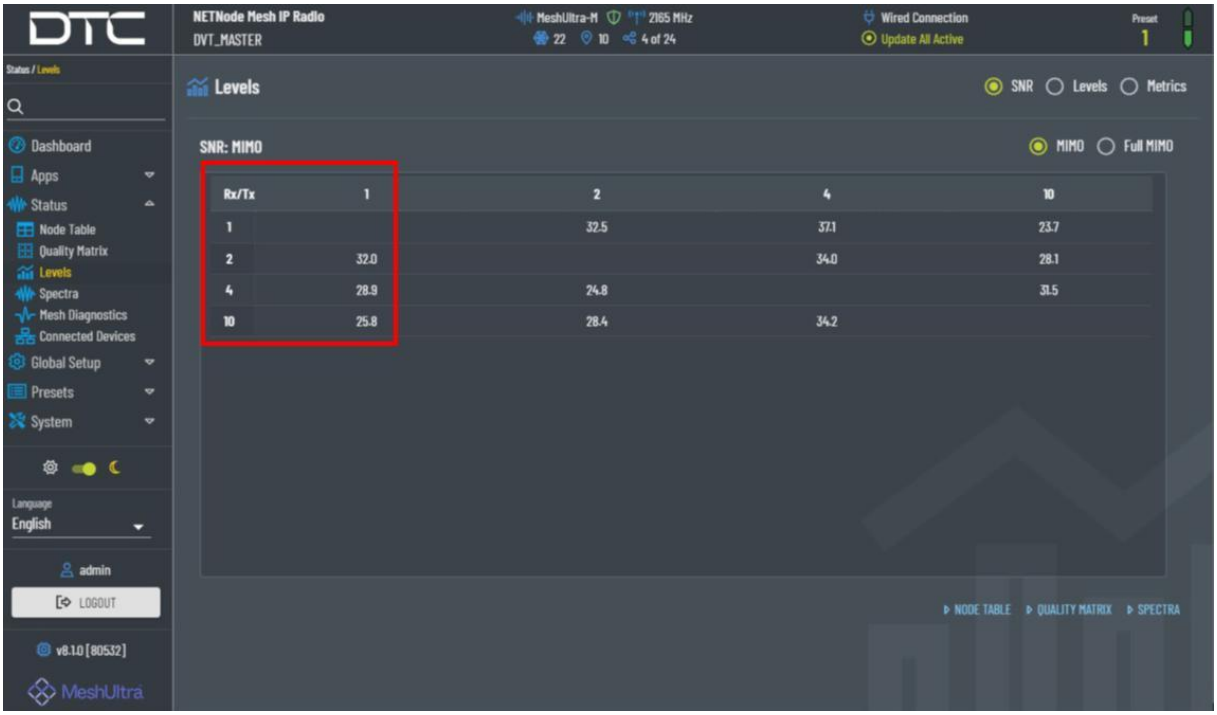


Figure 12. Levels screen, SNR view. The red box marks the values for Node 1.

6.3.1. How to read the table

- Rows are the **receiving** radio (Rx) and columns are the **transmitting** radio (Tx). Each cell is the value for the link between those two radios.
- **A higher number means better link quality.** In the Levels view the values are negative, so a value further above –100 (closer to zero) is better.
- A radio's value with itself (the diagonal) is not relevant. Ignore it.
- If a link is weak, act as you would for the Tactical Display: move the vehicle back, or improve the position of the relay drone.

6.3.2. Example (Figure 12)

In the column for Node 1, Node 2 receives Node 1 at **32.0 dB**, Node 4 at **28.9 dB** and Node 10 at **25.8 dB**. This means Node 1 has its strongest link with Node 2 and its weakest link with Node 10. The cell where Node 1 meets itself is ignored.

7. Quick Reference

7.1. Pre-Mission Checklist

	Step	Where
☐	Power on all radios: both vehicles, both Operator Boxes, main relay antenna, relay drone and GCS.	–
☐	Connect the ROCU-10 to the Operator Box and log in.	Section 3
☐	Confirm that every radio appears in the network.	Tactical Display / Node Table
☐	With the vehicles near the operator, run a spectrum scan.	Apps > Spectrum Scan
☐	If changing frequency: record the current frequency, set the new one on all radios, restart and verify.	Section 5.3
☐	Check that all links are green.	Status > Quality Matrix
☐	Check that no radio is above 20% occupancy.	Status > Node Table

7.2. Troubleshooting

Symptom	Likely cause	Action
Interface does not open in the browser	Wrong IP address or no wired connection	Check the cable and confirm the IP address for your configuration (Section 2.2).
Password rejected	A different password is set on this unit	Try the alternate password (Section 3.2).
Arrow turns orange or red during operation	Attenuation or range limit	Stop the advance, move the vehicle back, raise or reposition the relay drone.
A radio is missing after a frequency change	That radio was not set to the new frequency	Set the frequency on that radio manually (Section 5.3, Option B).
Communication is worse after a frequency change	The new frequency is not better	Return to the frequency you recorded before the change.
Network performance is slow	A radio has occupancy above 20%	Identify the radio in the Node Table.
Red or dark red cells in the Quality Matrix	Weak link between that pair of radios	Reposition the vehicle or the relay drone.

8. Glossary

Term	Definition
BLU-SDR	The Operator Box radio. Also the name of the Tactical configuration.
dB / dBm	Units for signal ratio (SNR, in dB) and absolute received power (Levels, in dBm).
GCS	Ground Control Station.
Mesh network	A network in which every radio can relay traffic for the others.
Node	Any radio participating in the Mesh network.
Occupancy	The share of bandwidth a node is using, in percent.
Phase-5	The GCS radio. Also the name of the Mesh configuration.
Preset	A saved set of radio settings.
Relay	Passing a signal on from one radio to another to extend range.
RF	Radio frequency.
ROCU-10	The operator control unit used to open the interface.
SNR	Signal-to-noise ratio. Higher is better.
Tx / Rx	Transmit / receive.